Convergence of single-valued fixed-point iteration for Gibbs-form equilibrium operators

ChatGPT (gpt-5.6-sol)*

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Abstract

For entropy-regularized time-inconsistent Markov decision processes, a regular relaxed equilibrium is characterized in Bayraktar, Huang, Wang, and Zhou as a fixed point of a single-valued Gibbs best-response operator Φ_λ. Existence follows from Brouwer's theorem, but convergence of the corresponding fixed-point iteration was left open.

We show that convergence does not hold under the original assumptions alone. A two-state model satisfying all of those assumptions has a unique regular relaxed equilibrium while the Picard iteration possesses a locally attracting two-cycle. Thus, even uniqueness of the equilibrium does not guarantee convergence.

We also give explicit, dimension-free sufficient conditions under which the auxiliary value operator Ψ_λ is ultimately contractive. Under these conditions the equilibrium is unique and both the value and policy iterations converge globally. For discounted rewards the conditions simplify substantially. Under exponential discounting, policy improvement yields unconditional geometric convergence for every $\lambda > 0$. Finally, we record the Jacobian of Ψ_λ and the associated local spectral-radius criterion.

1 The open problem

This paper addresses an open problem posed by Erhan Bayraktar, Yu-Jui Huang, Zhenhua Wang, Zhou Zhou in *Relaxed Equilibria for Time-Inconsistent Markov Decision Processes* [1] (Mathematics of Operations Research, 2024), Page 10, Remark 2.6 (end of Section 2.2).. The website catalogue entry is problem 136 (W4403683873_p0) of the Open Problems in Operations Research collection.

Give conditions under which the single-valued fixed-point iteration

$$y_{n+1} = \Psi_\lambda(y_n) \qquad (n \geq 0)$$

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converges (e.g., from arbitrary initialization, or from some specified set of initial points) to a fixed point $y^\star \in \mathbb{R}^d$ satisfying $\Psi_\lambda(y^\star) = y^\star$, equivalently producing a regular relaxed equilibrium $\pi^\star = \Gamma_\lambda(y^\star, \cdot)$ for the entropy-regularized time-inconsistent MDP.

The remainder of this document is the solution produced by the model, reproduced verbatim from the pipeline output.

Global convergence conditions, exponential discounting, and an attracting two-cycle Global convergence conditions, exponential discounting, and an attracting two-cycle

2 Introduction

Consider the entropy-regularized, infinite-horizon, discrete-time Markov decision process studied in [2]. Because the rewards may depend on the delay from the current time, the problem is generally time-inconsistent. The appropriate solution concept is consequently a subgame-perfect equilibrium among the agent's successive selves.

Entropy regularization makes the one-step best response unique. Accordingly, regular relaxed equilibria are fixed points of a single-valued Gibbs operator

$$\Phi_\lambda(\pi) = \Gamma_\lambda\big(V_\lambda^\pi, \cdot\big).$$

The existence theorem in [2] applies Brouwer's fixed-point theorem to an associated finite-dimensional operator. Brouwer's theorem proves existence, but it gives neither uniqueness nor convergence of the iteration

$$\pi_{n+1} = \Phi_\lambda(\pi_n).$$

The convergence of this iteration is the issue raised in [2, Remark 2.6].

The answer has two complementary parts.

1. Under explicit small-gain conditions, the auxiliary value iteration is globally convergent, and hence so is the policy iteration.
2. Without such additional conditions, convergence can fail: there is a two-state example with a unique equilibrium and a locally attracting period-two orbit.

The distinction between the policy operator Φ_λ and the value operator Ψ_λ is important. We prove below that their iterates are intertwined exactly, so a cycle of Ψ_λ produces a cycle of the operator Φ_λ appearing in the original convergence question.

3 Model and fixed-point operators

Let

$$\mathcal{S} = \{1, \dots, d\}$$

be the finite state space. Let $U \subset \mathbb{R}^\ell$ be compact with $0 < \mathrm{Leb}(U) < \infty$. An action $u \in U$ used at state i induces the transition vector

$$p_i^u = (p_{i1}^u, \dots, p_{id}^u) \in \mathcal{P}(\mathcal{S}).$$

Write

$$f_m(i,u) := f(m,i,u), \qquad m \in \mathbb{N}_0,$$

and let $(\delta_m)_{m\geq 0}$ be nonnegative with $\delta_0 = 1$.

For the estimates in Sections 3–5, it is convenient to use normalized volume

$$\nu(du) := \frac{du}{\mathrm{Leb}(U)}.$$

For $\mu \ll \nu$, define the entropy relative to ν by

$$H_\nu(\mu) := -\int_U \log\left(\frac{d\mu}{d\nu}\right) d\mu.$$

Thus

$$H_\nu(\mu) = -D_{\mathrm{KL}}(\mu\|\nu) \leq 0.$$

If instead H_{Leb} denotes differential entropy relative to Lebesgue measure, then

$$H_\nu(\mu) = H_{\mathrm{Leb}}(\mu) - \log \mathrm{Leb}(U).$$

Consequently, the original Lebesgue-entropy model is represented exactly in the normalized convention by replacing

$$f_m(i,u) \quad \text{with} \quad f_m(i,u) + \lambda\delta_m \log \mathrm{Leb}(U).$$

The added term is independent of i and u; in particular, its $m = 0$ contribution does not alter the Gibbs best response.

A regular relaxed policy is a collection $\pi = (\pi_i)_{i\in\mathcal{S}}$ of probability measures satisfying $\pi_i \ll \nu$ and $H_\nu(\pi_i) > -\infty$. Its shifted auxiliary value is

$$V_\lambda^\pi(i) = \mathbb{E}_i^\pi\left[\sum_{k=0}^{\infty}\left(f_{k+1}(X_k,U_k) + \lambda\delta_{k+1}H_\nu(\pi_{X_k})\right)\right].$$

The appearance of $k+1$, rather than k, is essential: this is precisely the one-step shift in the auxiliary value defined in [2, (2.9)].

For $y \in \mathbb{R}^d$, set

$$s_i^y(u) := f_0(i,u) + p_i^u \cdot y.$$

The unique entropy-regularized one-step best response is the Gibbs measure

$$\frac{d\Gamma_\lambda(y,i)}{d\nu}(u) = \frac{\exp(s_i^y(u)/\lambda)}{\displaystyle\int_U \exp(s_i^y(v)/\lambda)\,\nu(dv)}. \tag{1}$$

We write

$$\mu_i^y := \Gamma_\lambda(y,i).$$

The policy and value operators are

$$\Phi_\lambda(\pi) := \Gamma_\lambda(V_\lambda^\pi,\cdot), \qquad \Psi_\lambda(y) := V_\lambda^{\Gamma_\lambda(y,\cdot)}.$$

The Gibbs variational identity

$$\sup_{\mu\ll\nu}\left\{\int_U s_i^y(u)\,\mu(du) + \lambda H_\nu(\mu)\right\} = \lambda\log\int_U e^{s_i^y(u)/\lambda}\nu(du)$$

shows that the maximizer is unique and is given by (1). Hence a regular relaxed policy is an equilibrium if and only if it is a fixed point of Φ_λ.

3.1 Matrix representation

For $m \geq 1$, put

$$F_m := \max_{i \in \mathcal{S}} \sup_{u \in U} |f_m(i,u)| \,.$$

We impose the first-moment conditions

$$f_0 \text{ is bounded}, \qquad \sum_{m=1}^{\infty} mF_m < \infty, \qquad \sum_{m=1}^{\infty} m\delta_m < \infty. \tag{2}$$

Define

$$\begin{aligned} \omega_m &:= \max_{i \in \mathcal{S}} \sup_{u,v \in U} |f_m(i,u) - f_m(i,v)| \,, \\ \Delta &:= \max_{i \in \mathcal{S}} \sup_{u,v \in U} \|p_i^u - p_i^v\|_1 \,, \\ A_0 &:= \sum_{m=1}^{\infty} F_m, \qquad D_0 := \sum_{m=1}^{\infty} \delta_m, \qquad D_1 := \sum_{m=1}^{\infty} (m-1)\delta_m. \end{aligned} \tag{3}$$

Notice that $0 \leq \Delta \leq 2$ and $\omega_m \leq 2F_m$.

For the policy $\mu^y = (\mu_i^y)_i$, let

$$\begin{aligned} P_y(i,j) &:= \int_U p_{ij}^u \, \mu_i^y(du), \\ r_m^y(i) &:= \int_U f_m(i,u) \, \mu_i^y(du) + \lambda \delta_m H_\nu(\mu_i^y), \\ \Psi_\lambda(y) &= \sum_{k=0}^{\infty} P_y^k r_{k+1}^y. \end{aligned} \tag{4}$$

The last equality follows by conditioning on X_k. In particular, the reward vector accompanying P_y^k is r_{k+1}^y, in agreement with the shifted auxiliary value.

4 A global convergence theorem

For $R \geq 0$, define

$$\begin{aligned} Q(R) &:= \frac{1}{4\lambda} \left[\Delta^2 \sum_{m=1}^{\infty} (m-1)\big(F_m + \delta_m(\omega_0 + \Delta R)\big) + \Delta \sum_{m=1}^{\infty} \big(\omega_m + \delta_m(\omega_0 + \Delta R)\big) \right], \\ b &:= \Delta D_0 < 1, \qquad R_0 := \frac{A_0 + \omega_0 D_0}{1 - \Delta D_0}, \qquad Q(R_0) < 1. \end{aligned} \tag{5}$$

The series defining Q are finite by (2).

Theorem 1 (Global Picard convergence)**.** *Suppose* (2) *and* (5) *hold. Then:*

1. *Ψ_λ has a unique fixed point $y^\star \in B_{R_0} := \{y : \|y\|_\infty \leq R_0\}$;*
2. *for every $y_0 \in \mathbb{R}^d$, the Picard iteration*

$$y_{n+1} = \Psi_\lambda(y_n)$$

converges to $y^\star$;

3. Ψ_λ has no fixed point outside B_{R_0};

4. the corresponding regular relaxed equilibrium

$$\pi^\star = \Gamma_\lambda(y^\star, \cdot)$$

is unique;

5. for every initial regular relaxed policy π_0, the policy iteration

$$\pi_{n+1} = \Phi_\lambda(\pi_n)$$

converges statewise in total variation to $\pi^\star$.

4.1 Gibbs covariance identities

The proof rests on bounds that do not depend explicitly on the number of states.

For any bounded measurable $\varphi : U \to \mathbb{R}$ and $h \in \mathbb{R}^d$, differentiation of (1) gives

$$D\left(\int_U \varphi(u)\, \mu_i^y(du)\right)[h] = \frac{1}{\lambda} \operatorname{Cov}_{\mu_i^y}\big(\varphi(U), p_i^U \cdot h\big). \tag{6}$$

In particular,

$$\begin{aligned} DP_y(i,j)[h] &= \frac{1}{\lambda} \operatorname{Cov}_{\mu_i^y}\big(p_{ij}^U, p_i^U \cdot h\big), \\ \|DP_y(i,\cdot)[h]\|_1 &= \sup_{\|a\|_\infty \le 1} \frac{1}{\lambda} \left|\operatorname{Cov}_{\mu_i^y}\big(a \cdot p_i^U, p_i^U \cdot h\big)\right| \le \frac{\Delta^2}{4\lambda} \|h\|_\infty . \end{aligned} \tag{7}$$

Indeed, the oscillations of $a \cdot p_i^U$ and $p_i^U \cdot h$ are at most Δ and $\Delta \|h\|_\infty$, respectively. The last inequality then follows from

$$|\operatorname{Cov}(X,Y)| \le \sqrt{\operatorname{Var}(X)\operatorname{Var}(Y)} \le \frac{1}{4} \operatorname{osc}(X) \operatorname{osc}(Y).$$

Writing

$$Z_i(y) := \int_U e^{s_i^y(u)/\lambda} \nu(du),$$

we have

$$H_\nu(\mu_i^y) = \log Z_i(y) - \frac{1}{\lambda} \int_U s_i^y(u)\, \mu_i^y(du).$$

Therefore

$$\begin{aligned} DH_\nu(\mu_i^y)[h] &= -\frac{1}{\lambda^2} \operatorname{Cov}_{\mu_i^y}\big(s_i^y(U), p_i^U \cdot h\big), \\ Dr_m^y(i)[h] &= \frac{1}{\lambda} \operatorname{Cov}_{\mu_i^y}\big(f_m(i,U), p_i^U \cdot h\big) - \frac{\delta_m}{\lambda} \operatorname{Cov}_{\mu_i^y}\big(s_i^y(U), p_i^U \cdot h\big). \end{aligned} \tag{8}$$

4.2 Entropy and derivative bounds

If $\|y\|_\infty \le R$, then

$$\mathrm{osc}_{u\in U}\, s_i^y(u) \le \omega_0 + \Delta R.$$

Because $H_\nu = -D_{\mathrm{KL}}(\mu_i^y\|\nu) \le 0$, and because $\log Z_i(y)$ lies between the minimum and maximum of s_i^y/λ, we obtain

$$-\frac{\omega_0 + \Delta R}{\lambda} \le H_\nu(\mu_i^y) \le 0. \tag{9}$$

Consequently,

$$\begin{aligned} \|r_m^y\|_\infty &\le F_m + \delta_m(\omega_0 + \Delta R), \\ \|Dr_m^y[h]\|_\infty &\le \frac{\Delta}{4\lambda}\big(\omega_m + \delta_m(\omega_0 + \Delta R)\big)\, \|h\|_\infty . \end{aligned} \tag{10}$$

For the induced matrix norm, $\|P_y\|_\infty = 1$. Differentiating $P_y^k r_{k+1}^y$ gives

$$D(P_y^k)[h] = \sum_{\ell=0}^{k-1} P_y^\ell D P_y[h] P_y^{k-1-\ell}.$$

Equations (7) and (10), followed by summation over k, yield

$$\sup_{\|y\|_\infty \le R} \|D\Psi_\lambda(y)\|_{\infty\to\infty} \le Q(R), \qquad \|\Psi_\lambda(y) - \Psi_\lambda(z)\|_\infty \le Q(R)\, \|y - z\|_\infty \tag{11}$$

whenever $y, z \in B_R$. The first-moment assumptions justify termwise differentiation and uniform convergence of the differentiated series on every bounded ball.

The same estimates without differentiation give the global growth bound

$$\|\Psi_\lambda(y)\|_\infty \le A_0 + D_0(\omega_0 + \Delta\, \|y\|_\infty) = A_0 + \omega_0 D_0 + b\, \|y\|_\infty . \tag{12}$$

Proof of Theorem 1. By the definition of R_0,

$$A_0 + \omega_0 D_0 + bR_0 = R_0.$$

Thus (12) implies

$$\Psi_\lambda(B_{R_0}) \subseteq B_{R_0}.$$

By (11) and $Q(R_0) < 1$, Ψ_λ is a contraction on B_{R_0}. Banach's fixed-point theorem gives a unique fixed point $y^\star$ in this ball.

For an arbitrary initial point, let $R_n := \|y_n\|_\infty$. Equation (12) gives

$$R_n \le R_0 + b^n(R_0^{\mathrm{init}} - R_0), \qquad R_0^{\mathrm{init}} := \|y_0\|_\infty , \tag{13}$$

with the evident harmless modification when $R_0^{\mathrm{init}} < R_0$. Hence, for every $\varepsilon > 0$, the orbit eventually enters $B_{R_0+\varepsilon}$.

Because $Q(R)$ is continuous and $Q(R_0) < 1$, we may choose $\varepsilon > 0$ such that $Q(R_0 + \varepsilon) < 1$. Moreover,

$$\|y\|_\infty \le R_0 + \varepsilon \quad \Longrightarrow \quad \|\Psi_\lambda(y)\|_\infty \le R_0 + b\varepsilon < R_0 + \varepsilon.$$

Thus $B_{R_0+\varepsilon}$ is invariant and Ψ_λ is a contraction there. Every orbit therefore converges to $y^\star$ after entering this ball.

If y is any fixed point, (12) gives

$$\|y\|_\infty \le A_0 + \omega_0 D_0 + b\,\|y\|_\infty\,,$$

and hence $\|y\|_\infty \le R_0$. There are therefore no fixed points outside B_{R_0}.

The fixed-point correspondence shows that $\pi^\star = \Gamma_\lambda(y^\star,\cdot)$ is the unique regular relaxed equilibrium. Convergence of policy iteration follows from the intertwining identities proved in the next section and the total-variation continuity of $y \mapsto \Gamma_\lambda(y,i)$. □

5 The exact relation between Φ_λ and Ψ_λ

The value and policy operators satisfy

$$V_\lambda^{\Phi_\lambda(\pi)} = \Psi_\lambda(V_\lambda^\pi), \qquad \Phi_\lambda\big(\Gamma_\lambda(y,\cdot)\big) = \Gamma_\lambda\big(\Psi_\lambda(y),\cdot\big).$$

Indeed, both identities follow directly from $\Phi_\lambda(\pi) = \Gamma_\lambda(V_\lambda^\pi,\cdot)$ and $\Psi_\lambda(y) = V_\lambda^{\Gamma_\lambda(y,\cdot)}$.

In particular, if

$$\pi_{n+1} = \Phi_\lambda(\pi_n), \qquad y_n = V_\lambda^{\pi_n},$$

then

$$\pi_{n+1} = \Gamma_\lambda(y_n,\cdot), \qquad y_{n+1} = \Psi_\lambda(y_n). \tag{14}$$

Therefore convergence of the value iteration implies convergence of the induced policy iteration.

Likewise, if

$$\Psi_\lambda(y_A) = y_B, \qquad \Psi_\lambda(y_B) = y_A,$$

and

$$\pi_A = \Gamma_\lambda(y_A,\cdot), \qquad \pi_B = \Gamma_\lambda(y_B,\cdot),$$

then

$$\Phi_\lambda(\pi_A) = \pi_B, \qquad \Phi_\lambda(\pi_B) = \pi_A.$$

Thus a two-cycle for Ψ_λ is a genuine two-cycle for the policy operator Φ_λ appearing in the original open question.

6 Discounted rewards

Suppose now that

$$f_m(i,u) = \delta_m g(i,u), \qquad \delta_0 = 1,$$

and define

$$G := \max_i \sup_u |g(i,u)|\,, \qquad \omega := \max_i \operatorname{osc}_{u\in U} g(i,u).$$

Since the Gibbs score is

$$s_i^y(u) = g(i,u) + p_i^u \cdot y,$$

define

$$\bar r^y(i) := \int_U g(i,u)\,\mu_i^y(du) + \lambda H_\nu(\mu_i^y), \qquad r_m^y = \delta_m \bar r^y, \qquad \Psi_\lambda(y) = \sum_{k=0}^\infty \delta_{k+1} P_y^k \bar r^y. \tag{15}$$

There is a useful cancellation in the derivative. Equations (6) and (8) give

$$\begin{aligned} D\bar{r}^y(i)[h] &= -\frac{1}{\lambda}\operatorname{Cov}_{\mu_i^y}\big(p_i^U\cdot y, p_i^U\cdot h\big), \\ Dr_m^y(i)[h] &= -\frac{\delta_m}{\lambda}\operatorname{Cov}_{\mu_i^y}\big(p_i^U\cdot y, p_i^U\cdot h\big). \end{aligned} \tag{16}$$

Thus the first line is the derivative of $\bar{r}^y$, whereas the second is the derivative of r_m^y.

If $\|y\|_\infty \le R$, then

$$\|\bar{r}^y\|_\infty \le G+\omega+\Delta R, \qquad \|D\bar{r}^y\|_{\infty\to\infty} \le \frac{\Delta^2 R}{4\lambda}.$$

Consequently,

$$\sup_{\|y\|_\infty \le R} \|D\Psi_\lambda(y)\|_{\infty\to\infty} \le Q_{\text{disc}}(R) := \frac{\Delta^2}{4\lambda}\left[D_1(G+\omega+\Delta R)+D_0 R\right]. \tag{17}$$

Furthermore,

$$\|\Psi_\lambda(y)\|_\infty \le D_0(G+\omega+\Delta\,\|y\|_\infty).$$

It follows that the discounted specialization of Theorem 1 is obtained under

$$\boxed{\Delta D_0 < 1, \qquad R_{\text{disc}} := \frac{D_0(G+\omega)}{1-\Delta D_0}, \qquad \frac{\Delta^2}{4\lambda}\left[D_1(G+\omega+\Delta R_{\text{disc}})+D_0 R_{\text{disc}}\right] < 1.} \tag{18}$$

Corollary 1. *Under (2) and (18), the discounted problem has a unique regular relaxed equilibrium, and both the value and policy Picard iterations converge globally.*

7 Unconditional convergence under exponential discounting

The exponential case admits a stronger result that does not require (18). Suppose

$$\delta_m = \beta^m, \qquad f_m(i,u) = \beta^m g(i,u), \qquad 0<\beta<1.$$

For a regular relaxed policy π, explicitly define

$$r_i^\pi := \int_U g(i,u)\,\pi_i(du) + \lambda H_\nu(\pi_i).$$

Let P^π be its transition matrix, and define

$$T^\pi w := r^\pi + \beta P^\pi w.$$

The entropy-regularized Bellman operator is

$$\begin{aligned} (Tw)_i &= \sup_{\mu\ll\nu}\left\{\int_U \big(g(i,u)+\beta p_i^u\cdot w\big)\mu(du) + \lambda H_\nu(\mu)\right\} \\ &= \lambda \log \int_U \exp\left(\frac{g(i,u)+\beta p_i^u \cdot w}{\lambda}\right)\nu(du). \end{aligned}$$

It is monotone and is a β-contraction in the sup norm.

Let

$$w^\pi = (I - \beta P^\pi)^{-1} r^\pi.$$

The shifted auxiliary value is

$$V_\lambda^\pi = \beta w^\pi.$$

Hence policy iteration takes the familiar soft policy-improvement form

$$\pi_{n+1} = \Gamma_\lambda(\beta w^{\pi_n}, \cdot).$$

By construction,

$$T^{\pi_{n+1}} w^{\pi_n} = T w^{\pi_n} \geq T^{\pi_n} w^{\pi_n} = w^{\pi_n}.$$

Monotone policy evaluation therefore yields

$$w^{\pi_{n+1}} \geq T w^{\pi_n} \geq w^{\pi_n}.$$

Let $w^\star$ be the unique fixed point of T. Every policy value satisfies $w^\pi \leq w^\star$. It follows componentwise that

$$0 \leq w^\star - w^{\pi_{n+1}} \leq T w^\star - T w^{\pi_n}, \qquad \left\| w^\star - w^{\pi_{n+1}} \right\|_\infty \leq \beta \left\| w^\star - w^{\pi_n} \right\|_\infty. \tag{19}$$

Theorem 2 (Exponential discounting)**.** *For exponential discounting, every $\lambda > 0$, and every initial regular relaxed policy π_0,*

$$\left\| w^\star - w^{\pi_n} \right\|_\infty \leq \beta^n \left\| w^\star - w^{\pi_0} \right\|_\infty.$$

Moreover,

$$\pi_n \longrightarrow \pi^\star := \Gamma_\lambda(\beta w^\star, \cdot)$$

statewise in total variation. In particular, the regular relaxed equilibrium is unique and policy iteration converges globally.

Proof. The value estimate is (19) iterated n times. The Gibbs formula shows that $y \mapsto \Gamma_\lambda(y, i)$ is continuous in total variation. Since $\beta w^{\pi_n} \to \beta w^\star$, the policies converge to $\pi^\star$. □

8 A two-state counterexample

We now show that convergence can fail for nonexponential data even when the equilibrium is unique. In this section only, entropy is taken relative to Lebesgue measure, exactly as in [2]:

$$H_{\mathrm{Leb}}(\rho) = -\int_U \rho(u) \log \rho(u)\, du.$$

Let

$$\mathcal{S} = \{1, 2\}, \qquad U = A \cup B, \qquad A = [-2, -1], \qquad B = [1, 2].$$

Both components have Lebesgue measure one and are separated by distance two. Set

$$\lambda = \frac{1}{\log 3}, \qquad \delta_0 = \delta_1 = 1, \qquad \delta_m = 0 \quad (m \geq 2).$$

For actions belonging to the two components, define the data by

state i	action component	p_i^u	$f_0(i,u)$	$f_1(i,u)$
1	A	$(1,0)$	0	0
1	B	$(1,0)$	-1	0
2	A	$(1,0)$	0	6
2	B	$(0,1)$	-4	2

$$, \qquad f_m \equiv 0 \quad (m \geq 2). \tag{20}$$

These data satisfy the assumptions in [2]:

- U is compact and has positive Lebesgue measure.
- The summability assumption holds because the rewards and discount sequence have finite support.
- The rewards and transition probabilities are piecewise constant. They are Lipschitz on U because the two components are separated by distance two.
- In dimension one, the uniform cone condition holds with, for example, $\vartheta = 1/2$: from every point of either component, a one-sided interval of length $1/2$ contained in that component can be chosen.

Let q denote the Gibbs probability assigned to the component B. At state 1 the transition vector is independent of the action, while the score on B is one unit smaller than the score on A. Since $1/\lambda = \log 3$, the Gibbs mass of B is

$$\frac{e^{-1/\lambda}}{1+e^{-1/\lambda}} = \frac{1/3}{1+1/3} = \frac{1}{4}.$$

At state 2, write

$$x := y_2 - y_1.$$

The score on A is y_1, while the score on B is $-4 + y_2$. Thus

$$q(x) = \frac{e^{(x-4)/\lambda}}{1+e^{(x-4)/\lambda}} = \frac{3^{x-4}}{1+3^{x-4}}.$$

Let

$$h(q) := -q\log q - (1-q)\log(1-q), \qquad c := \lambda h\left(\frac{1}{4}\right).$$

Because both action components have length one, the Lebesgue-relative entropy of a policy assigning mass q to B is exactly $h(q)$. Since only the $m=1$ term appears in the auxiliary value, the value operator is

$$\Psi_\lambda(y_1, y_2) = \big(c,\, 6 - 4q(y_2-y_1) + \lambda h\big(q(y_2-y_1)\big)\big)\,. \tag{21}$$

Now

$$q(3) = \frac{1}{4}, \qquad q(5) = \frac{3}{4}, \qquad h\left(\frac{1}{4}\right) = h\left(\frac{3}{4}\right).$$

It follows immediately that, with

$$y_A := (c, c+3), \qquad y_B := (c, c+5),$$

and

$$\pi_A := \Gamma_\lambda(y_A, \cdot), \qquad \pi_B := \Gamma_\lambda(y_B, \cdot),$$

we have

$$\Psi_\lambda(y_A) = y_B, \qquad \Psi_\lambda(y_B) = y_A, \qquad \Phi_\lambda(\pi_A) = \pi_B, \qquad \Phi_\lambda(\pi_B) = \pi_A. \tag{22}$$

Thus the policy iteration named in the original convergence question fails to converge.

8.1 Uniqueness of the fixed point

After one iteration, the first coordinate is c. Therefore a fixed point must have the form $(c, c+x)$, where

$$x = \varphi(x), \qquad \varphi(x) := 6 - 4q(x) + \lambda h(q(x)) - c.$$

Because

$$q'(x) = (\log 3)q(x)(1 - q(x))$$

and

$$h'(q(x)) = \log \frac{1 - q(x)}{q(x)} = -(x - 4)\log 3,$$

the choice $\lambda = 1/\log 3$ gives

$$\varphi'(x) = -x(\log 3)q(x)(1 - q(x)).$$

Moreover,

$$c \le \lambda \log 2 = \frac{\log 2}{\log 3} < 1,$$

so

$$\varphi(x) \ge 2 - c > 0.$$

Hence $\varphi(x) - x > 0$ for $x \le 0$. On $(0, \infty)$,

$$\frac{d}{dx}\big(\varphi(x) - x\big) = -x(\log 3)q(x)(1 - q(x)) - 1 < 0.$$

Finally,

$$\varphi(x) - x \longrightarrow -\infty \qquad \text{as } x \to \infty.$$

The equation $\varphi(x) = x$ therefore has exactly one solution. Consequently, Ψ_λ has a unique fixed point, and the fixed-point correspondence implies that Φ_λ has a unique regular relaxed equilibrium.

Thus uniqueness alone does not ensure convergence of policy iteration.

8.2 Local attraction of the cycle

At the two cycle points,

$$\varphi'(3) = -\frac{9\log 3}{16}, \qquad \varphi'(5) = -\frac{15\log 3}{16}.$$

The multiplier of the second iterate is therefore

$$\big|(\varphi^2)'(3)\big| = \big|\varphi'(5)\varphi'(3)\big| = \frac{135(\log 3)^2}{256} \approx 0.636477 < 1.$$

Hence the two-cycle is locally attracting. Since Ψ_λ maps all of $\mathbb{R}^2$ onto the line $\{(c, c+x) : x \in \mathbb{R}\}$ in one step, this is also local attraction for the full two-dimensional value iteration.

8.3 Effect of normalized entropy

For this example, $\mathrm{Leb}(U) = 2$, so

$$H_\nu(\mu) = H_{\mathrm{Leb}}(\mu) - \log 2.$$

If the rewards in (20) are kept fixed and the entropy convention is changed to normalized volume, set

$$\kappa := \lambda \log 2.$$

The Gibbs policies are unchanged, while the auxiliary value is translated by $-\kappa\mathbf{1}$. Because

$$\Gamma_\lambda(y - a\mathbf{1}, \cdot) = \Gamma_\lambda(y, \cdot)$$

for every scalar a, the translated cycle is

$$(c - \kappa, c - \kappa + 3) \longleftrightarrow (c - \kappa, c - \kappa + 5).$$

Thus the normalization changes only a constant translation; it does not remove the counterexample or alter its stability.

9 A local spectral-radius criterion

The preceding global sufficient conditions are deliberately explicit. Near a particular fixed point, the exact Jacobian gives a sharper criterion.

Let $y^\star$ be a fixed point and abbreviate

$$P_\star := P_{y^\star}, \qquad r_m^\star := r_m^{y^\star}.$$

For a direction $h \in \mathbb{R}^d$, differentiation of (4) yields

$$\boxed{D\Psi_\lambda(y^\star)[h] = \sum_{k=0}^{\infty} \left[P_\star^k D r_{k+1}^{y^\star}[h] + \sum_{\ell=0}^{k-1} P_\star^\ell D P_{y^\star}[h] P_\star^{k-1-\ell} r_{k+1}^\star \right],} \tag{23}$$

where the inner sum is empty for $k = 0$. The first-moment assumptions (2), together with (7) and (10), imply absolute convergence of this series in operator norm.

Proposition 1 (Local linearization criterion)**.** *Let*

$$J := D\Psi_\lambda(y^\star).$$

Then:

1. *if $\rho(J) < 1$, the fixed point $y^\star$ is locally asymptotically stable;*
2. *if $\rho(J) > 1$, the fixed point is unstable;*
3. *if $\rho(J) = 1$, linearization alone is inconclusive.*

Proof. If $\rho(J) < 1$, finite-dimensional spectral theory provides an equivalent norm for which the induced norm of J is strictly less than one. Continuity of $D\Psi_\lambda$ then makes Ψ_λ a contraction on a sufficiently small neighborhood of $y^\star$.

If $\rho(J) > 1$, the standard discrete-time linearized instability theorem applies. Eigenvalues on the unit circle are the nonhyperbolic case and require higher-order analysis. $\square$

This criterion is intrinsically a statement about the finite-dimensional value operator Ψ_λ. Along an induced value iteration, continuity of Γ_λ transfers value convergence to policy convergence in total variation. It should not, however, be interpreted as a spectral criterion on the entire policy space unless a policy topology and a corresponding differentiable structure are specified.

10 Conclusion

The fixed-point iteration proposed for computing regular relaxed equilibria has a precise dichotomy.

First, explicit dimension-free estimates yield global convergence when the transition sensitivity, discount tail, and inverse temperature satisfy the small-gain conditions (5). The proof gives an invariant radius, eventual entrance into a contractive ball, uniqueness of the equilibrium, and exclusion of fixed points outside that ball. Discounted rewards admit the sharper conditions (18), while exponential discounting gives unconditional geometric convergence through soft policy improvement.

Second, convergence is false under the original assumptions alone. The example in Section 6 satisfies all of those assumptions, possesses a unique regular relaxed equilibrium, and nevertheless has a locally attracting two-cycle for both Ψ_λ and the exact policy operator Φ_λ. Brouwer's existence theorem therefore cannot be strengthened to an unconditional convergence claim.

References

[1] Erhan Bayraktar, Yu-Jui Huang, Zhenhua Wang, Zhou Zhou. *Relaxed Equilibria for Time-Inconsistent Markov Decision Processes.* Mathematics of Operations Research, 2024. `https://doi.org/10.1287/moor.2023.0209`

[2] E. Bayraktar, Y.-J. Huang, Z. Wang, and Z. Zhou, *Relaxed Equilibria for Time-Inconsistent Markov Decision Processes*, Mathematics of Operations Research **50** (2025), no. 4, 2666–2687. doi:10.1287/moor.2023.0209; arXiv:2307.04227.